INTELLIGENT SONAR BASED OBJECT CLASSIFICATION SYSTEM

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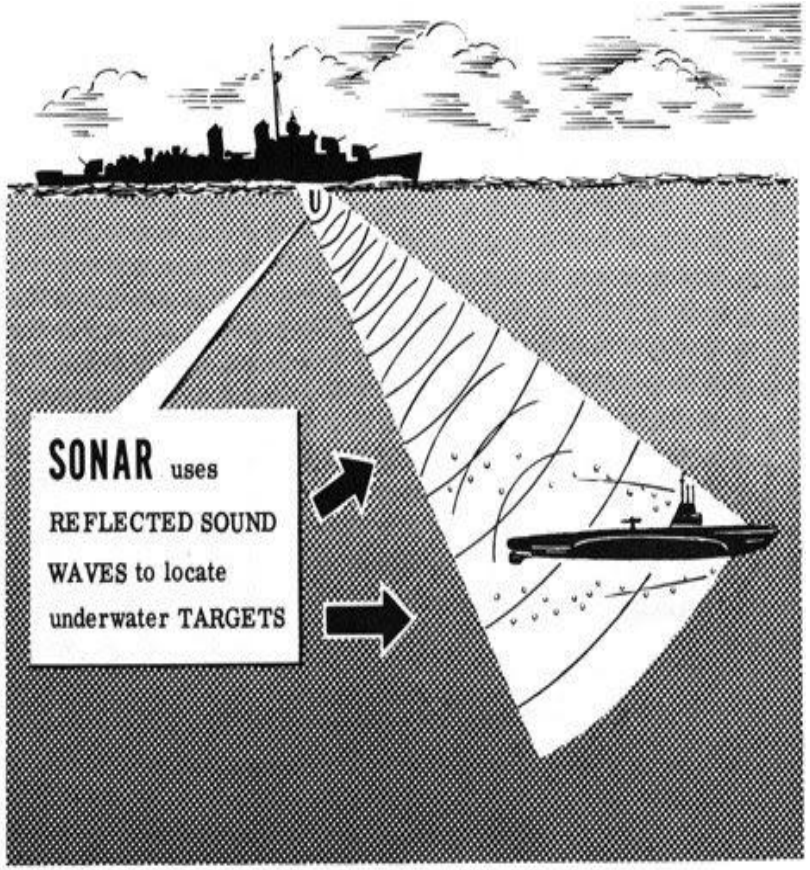
ABSTRACT

This project presents an intelligent SONAR-based system that uses ultrasonic sensing and machine learning to classify object behavior. Instead of only measuring distance, the system analyzes reflected ultrasonic signals to classify detected objects as hard-like or soft-like under controlled, short-range conditions. Real-time data is collected using an Arduino-based setup and processed using a supervised learning model with confidence-based decision logic to reduce false alarms. The system also includes a simple AI-based voice assistant that provides real-time spoken updates about detected object behavior. Overall, the project demonstrates how low-cost ultrasonic sensors can be enhanced with intelligence and voice-based reporting for surveillance and monitoring applications in controlled environments.



Existing Systems & Drawbacks

- Conventional ultrasonic and basic SONAR systems are mainly used for **object detection and distance measurement**.
- These systems only indicate that an object is present, without providing any additional understanding.
- Monitoring and interpretation of detected objects are mostly done **manually by humans**.
- Many low-cost systems do not provide any form of intelligent decision support.
- Cannot differentiate between different types of object behavior.
- High chance of false alarms due to lack of intelligent analysis.
- No automatic alert or voice-based reporting mechanism.

Module description

1. Hardware Setup

Ultrasonic sensors , servo motor and an Arduino are used to detect nearby objects.

2. Data Collection & Model Training

Collect real-time data . collected ultrasonic data is processed and used to train a machine learning model to classify object behavior.

3. AI Voice Assistant

The system provides real-time voice alerts to report detected object behavior without any human intervention .

4. UI Integration

A simple user interface displays live detection ,classification results and interacting with Voice assistant.